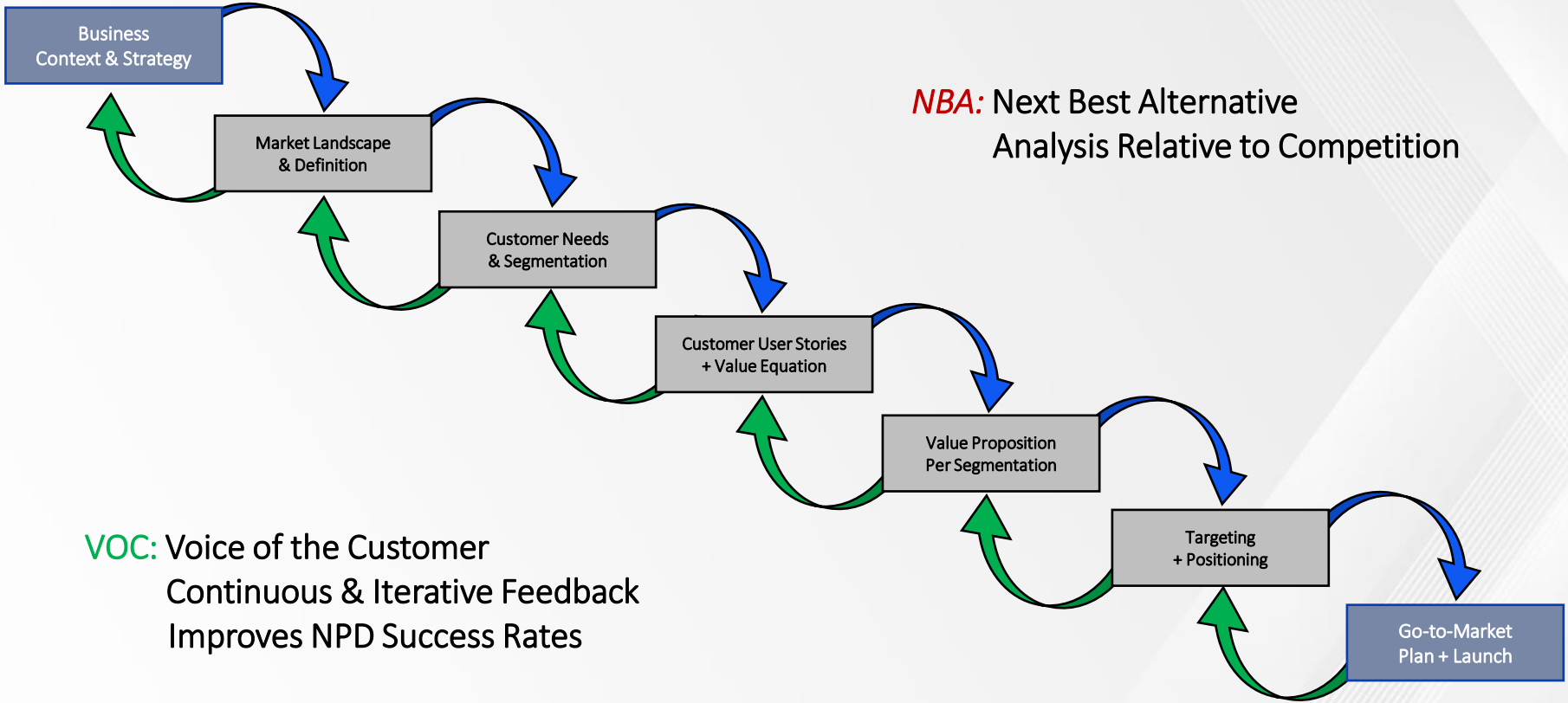
# *New Product Strategy:* Addresses Risks vs Competitive NBA




*NBA:* Next Best Alternative
Analysis Relative to Competition

VOC: Voice of the Customer
Continuous & Iterative Feedback
Improves NPD Success Rates

# Robust Stage-Gate: Collaborative Input Improves NPD Success

Development Stages

Iteration

POC / MVP / Future Versions

| Concept | Plan | Develop | Qualify | Launch | Maximize | Retire / EOL |
|---|---|---|---|---|---|---|
| G1 | G2 | G3 | G4 | G5 | G6 | G7 |

**Understand Needs**

- Understand Customer Needs
- Versus Next Best Alternative




**Validate Concepts**

- Refine and validate value proposition and segmentation




**Test Prototypes with Customers**

- Iteratively validate evolving offering
- Validate final design

**Evaluate Offering and Adapt**

- Assess success and adapt to marketplace

# Project Nest-AI

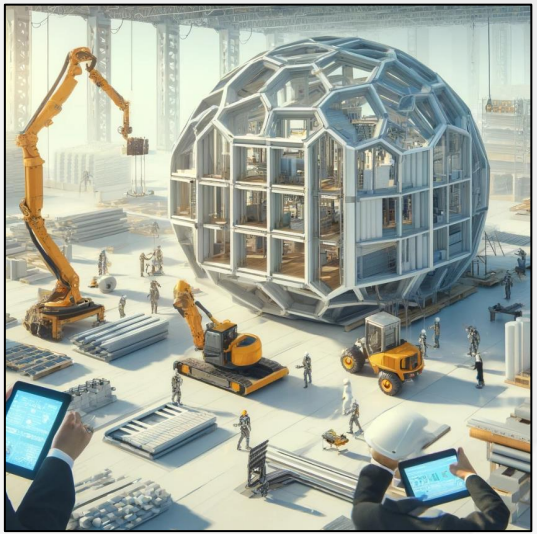
## Cloud & Edge AI Home Building Platform

### Today



- Extended permitting and approval processes slow development.
- Minimal design customization available for buyers.
- Predominantly traditional, time-consuming construction methods.

### Future



- AI-enhanced permitting streamlines approval processes.
- Home buyers enjoy extensive customization options.
- Prefabricated construction methods boost efficiency.

Research/Demo

Stage: *Planning*





# Customer CONOPS: *Project Nest - AI*

**Business:** Nest AI **Project #:** 24.001A

## CONOPS



| | |
|---|---|
| 1 | **Buying Lots:** Homebuilders purchase premium lots from developers. |
| 2 | **Offering Plans:** Standardized floor plans presented to buyers. |
| 3 | **Revising Designs:** Costly, repeated design revisions managed efficiently. |
| 4 | **Simplifying Permits:** Streamlining complex permitting processes with AI. |
| 5 | **Optimizing Construction**: Reducing time and waste in construction phases. |
| 6 | **Easing Closure:** Facilitating stress-free home buying closures. |
| Future Capabilities | **Enhancing Customization:** Empowering buyers with advanced 3D home customization tools. |
| Future Capabilities | **Promoting Sustainability:** Advocating for eco-friendly building practices and materials. |

Research/Demo

Stage: *Planning*





# Work - Flow by Persona: *Project Nest - AI*

**Business:** Nest AI **Project #:** 24.001A

## Personas

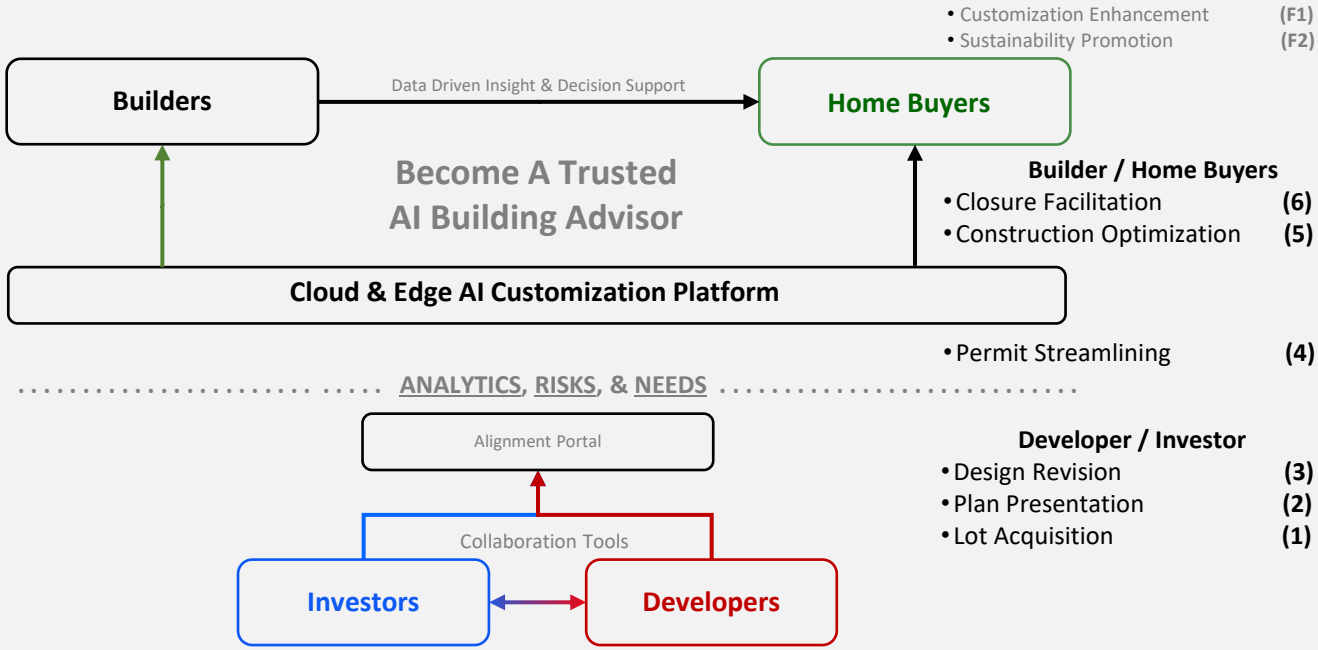
**Investors:**
- Financially driven, embracing tech for rapid, profitable community development

**Developers:**
- Financially driven, embracing tech for rapid, profitable community development.

**Builder:**
- Efficiency-focused, integrating AI and prefab for quality construction.

**Home Buyers:**
- Desiring affordable, customizable, and eco-conscious homes.

## Customer + Decision Work-Flow




**Desired Outcome:** Profitably creating personalized, efficient, sustainable communities.

Research/Demo

Stage: *Planning*

# Customer Pain Points: *Project Nest - AI*

**Business:** Nest AI **Project #:** 24.001A





## Personas

**Investors:**
- Financially driven, embracing tech for rapid, profitable community development

**Developers:**
- Financially driven, embracing tech for rapid, profitable community development.

**Builder:**
- Efficiency-focused, integrating AI and prefab for quality construction.

**Home Buyers:**
- Desiring affordable, customizable, and eco-conscious homes.

## Pain Points

**Investors:**
- Navigating financial risks in volatile markets.
- Finding investments with strong, stable returns.
- Ensuring projects meet stringent ESG criteria.

**Developers:**
- High upfront land and development costs.
- Delays in permitting hinder project timelines.
- Adapting to tech advancements can be challenging.

**Builder:**
- Construction delays impact project deliverables.
- Managing increased demand for sustainable materials.
- Integrating AI with existing construction practices.

**Home Buyers:**
- Overwhelming costs limit housing affordability.
- Limited options for home personalization.
- Eco-friendly homes often come at a premium.



**Note:** Example Only, Modify as Needed

**Reference Links:** Doc 1, Doc 2, Doc 3

Research/Demo

Stage: *Planning*





# Customer Problem Statement: *Project Nest - AI*

**Business:** Nest AI **Project #:** 24.001A

| | | | |
|---|---|---|---|
| 1 | Problem Statement | - | The traditional homebuilding process is inefficient, environmentally harmful, and lacks customization, leading to dissatisfied stakeholders. |
| 2 | Customer Outcomes | - | Homes crafted to buyer's personal tastes. |
| | | - | Sustainable living spaces, lower environmental impact. |
| | | - | Streamlined buying process, less stress involved. |
| 3 | Customer Why's | - | Personalized home design reflecting individual lifestyle needs. |
| | | - | Commitment to eco-friendly, sustainable living practices. |
| | | - | Streamlined, hassle-free home purchasing experience desired. |
| 4 | Problems with the Status Quo | - | Home customization options are too restrictive. |
| | | - | Construction processes often result in resource waste. |
| | | - | Home buying experience fraught with complexities, stress. |



DE-RISKING STRATEGIES

# 03

# Portfolio Summary Templates

Executive Summary + Detailed Deliverables for Review

# Nest - AI

| COGS | MSRP | Mgn % |
|---|---|---|
| $ -- k | $ -- k | % |



| Functional Lead | Name |
|---|---|
| Product Mgr | **A. Seguin** |
| Project Eng | Eco-Spheres |
| BD/Sales Lead | Partner |

| COGS | To-Date | 2023 | Life |
|---|---|---|---|
| I-RAD | -- | -- | -- |
| C-RAD | -- | -- | -- |
| Cap-Ex | -- | -- | -- |

| Financials | |
|---|---|
| 1st Rev Date | Q# - YY |
| 3-Year NPV | $ -- M |
| IRR | % |

Research/Demo

## Project Overview

### Project Summary

- ***Nest AI*** is a cutting-edge homebuilding enhancement platform that incorporates artificial intelligence to revolutionize the home design and construction process. Our platform empowers homebuyers with tools for personalizing their living spaces and facilitates seamless communication between developers, builders, and buyers.

- Scheduled for its Beta Launch in 2024, ***Nest AI*** is set to introduce intelligent design interfaces and advanced prefab construction protocols. The project will roll out a range of features.

## Market Opportunity

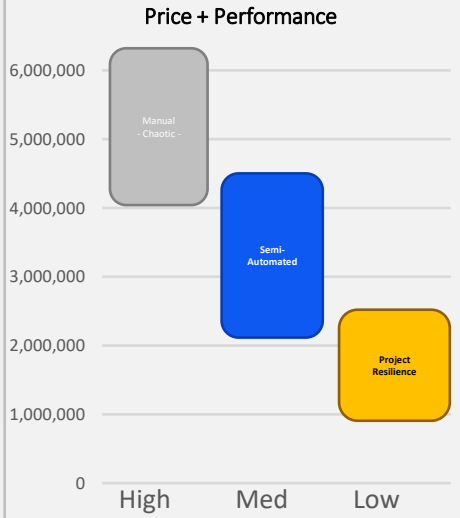
| Franchise Pursuits | | |
|---|---|---|
| **Pursuit** | **Award** | **Full value** |
| Prog 1 | YYYY | $ - M |
| Prog 2 | YYYY | $ - M |
| Prog 3 | YYYY | $ - M |

| CRM: Dynamics | | |
|---|---|---|
| **Date** | **Award** | **Full value** |
| Q# - YY | YYYY | $ - M |
| Q# - YY | YYYY | $ - M |
| Q# - YY | YYYY | $ - M |
| Q# - YY | YYYY | $ - M |

Comments
- X
- X
- X
- X

## Portfolio Positioning

| Value Creation | Value Capture |
|---|---|
| $ 2 M | $ 1 M |




## Product Strategy / Value Proposition

**Product Strategy**
- **Introduce Nest AI:** Home Customization and Building Platform.
- **Beta Launch:** Interactive Design Interface and Prefab Protocols.
- **Operational Spectrum:** From Design Tools to AI Project Management.

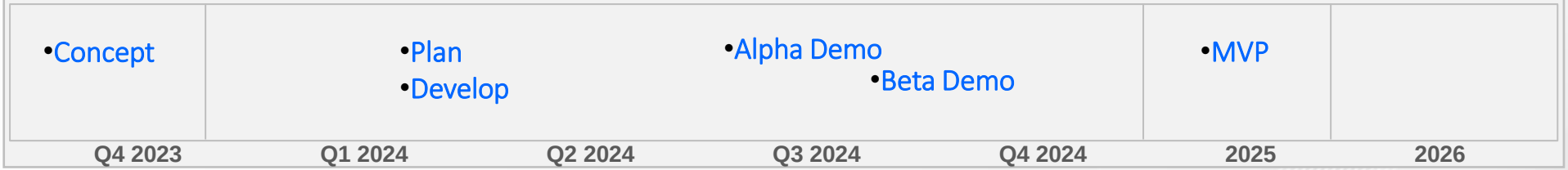
**Value Proposition**
- Customized home design AI.
- Real-time construction updates.
- Scalable homebuilding solutions.
- Efficient project management systems.
- 3D interactive design tools.
- Sustainable building practices integration.

## Dependencies: Intra-BU/LOB + Customer

| Eco-Spheres | |
|---|---|
| **Project** | **Project #** |
| Prog 1 | YYYY |
| Prog 2 | YYYY |
| Prog 3 | YYYY |

| Veteran - Customer | |
|---|---|
| **Project** | **Project #** |
| Prog 1 | YYYY |
| Prog 2 | YYYY |
| Prog 3 | YYYY |

## Roadmap + Schedule






Reference Links: Doc 1, Doc 2, Doc 3

DE-RISKING STRATEGIES

# Nest AI



**Business:** Nest AI **Project #:** 24.004W

Research/Demo

## Project Description

**Project Summary**

- ***Nest AI*** is a cutting-edge homebuilding enhancement platform that incorporates artificial intelligence to revolutionize the home design and construction process. Our platform empowers homebuyers with tools for personalizing their living spaces and facilitates seamless communication between developers, builders, and buyers.

## Information

**Segment**

- *X*

| Functional Lead | Name |
|---|---|
| Product Mgr | **A. Seguin** |
| Project Eng | AUTMHQ |
| BD/Sales Lead | F. Last |

**Product Variant 1**

| COGS | MSRP | Mgn % |
|---|---|---|
| $ -- k | $ -- k | % |

**Product Variant 2**

| COGS | MSRP | Mgn % |
|---|---|---|
| $ -- k | $ -- k | % |



## Objectives

**2023**

- Concept

**2024 + Beyond**

- Plan / Develop
- Alpha / Beta Demo
- MVP System

**Related Activities**

- User Test/Feedback
- Partnership Development

## Dependencies

**Eco-Spheres :** Internal I-RAD

- X
- Y
- Z

**Eco-Spheres:** External C-RAD

- X
- Y
- Z

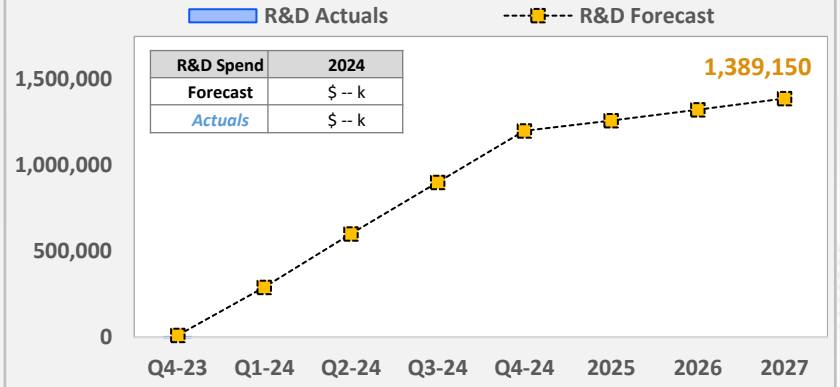
**Eco-Spheres :** Customer

- X
- Y
- Z

## Critical Issues & Risks

| Risk | Issue | Counter Measure / Mitigation | Owner |
|---|---|---|---|
| C1 | X | X | F. Last |
| C2 | X | X | F. Last |
| M1 | X | X | F. Last |
| M2 | X | X | F. Last |
| M3 | X | X | F. Last |
| M4 | X | X | F. Last |
| M5 | X | X | F. Last |

## Project Cost Actuals vs Forecast: k$




## Roadmap + Schedule




Reference Links: Doc 1, Doc 2, Doc 3



DE-RISKING STRATEGIES

# Platform Eco-System: Cloud & Edge AI Home Building Platform



Experienced Construction + AI Technology Leadership Teams

Home Buyers demanding fast, cost-effective designs; focus on green living.

**Hardware + Annual Subscription**

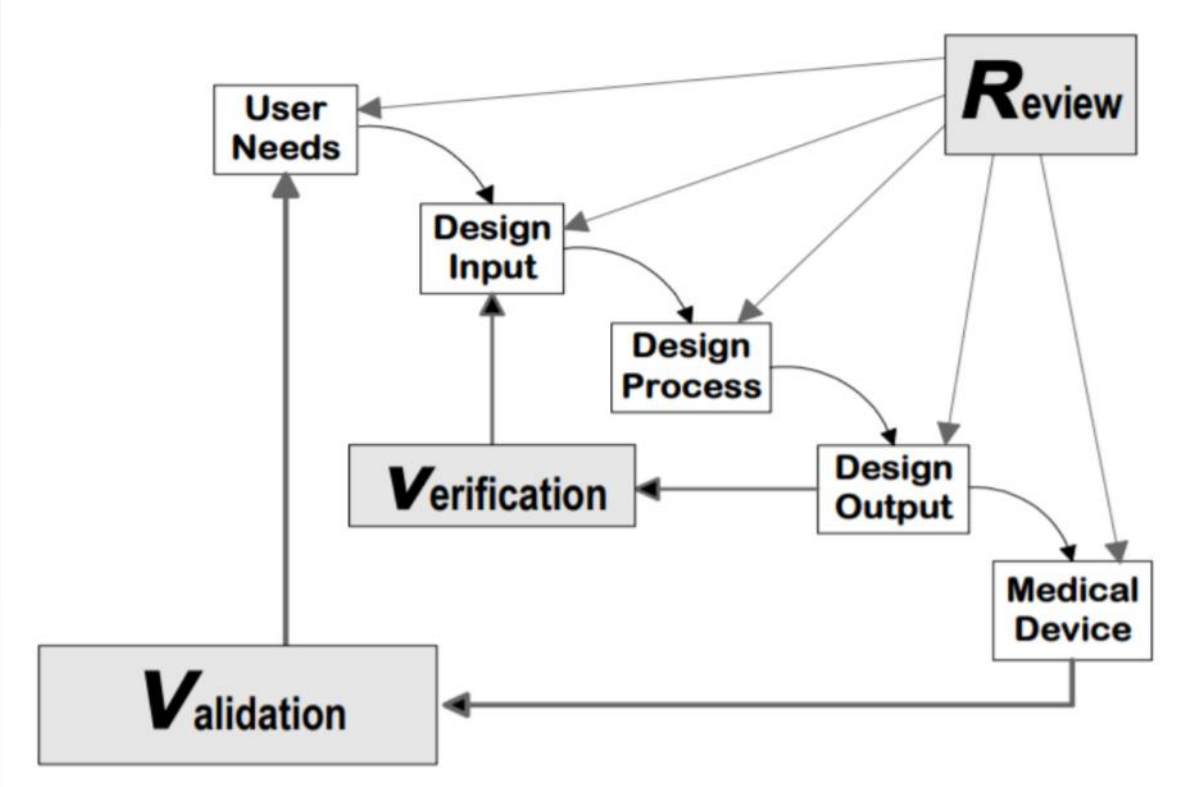
Investors seeking quick, sustainable returns; Builders prioritizing AI, efficiency, speed.

AI/ML development teams & software collaboration partners

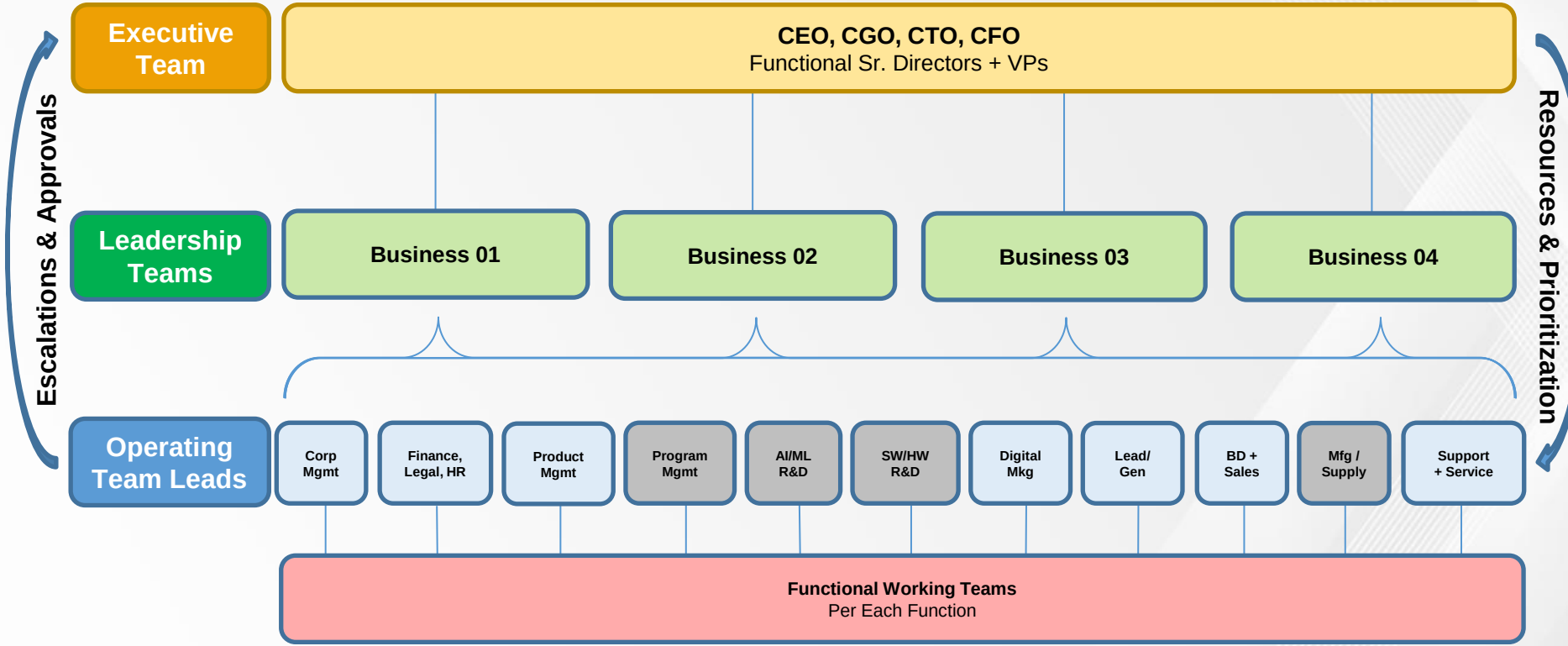
# Index

## Back-Up Templates

# Concepts: *FDA References*



# Cross-Functional Virtual Teams: *Escalations, Approvals, & Prioritization*



# Product Requirements: *Design Traceability Matrix*

Plan Stage
- Customer User Stories
- Customer Requirement Specs

Develop Stage: Design Matrix
- Team Specification #
- Design Output #

| Dev Team | Design Input # | Specification Input: Target at Launch | Market Input: Stretch Target | Design Output # | Design Output: Definable / Measurable |
|---|---|---|---|---|---|
| Software | CRS-01.IN.SRS.001 | Must be able to ***upload/delete/re-upload video versions*** to feed customer portal website. | Enable Admin User Log-In/Password to perform upload. | CRS-01.OUT.SRS.001 | Upload/Delete 4 videos in < 5 min on 100 MB/s connection. |
| Software | CRS-02.IN.SRS.002 | Doctor office/Lab to provide QR code (encoded format internally) and/or unique KEY to ***automate patient account creation.*** | Unique KEY expires between 1-7 days (Admin may change) | CRS-02.OUT.SRS.002 | |

Qualify Stage: Design Matrix
- Design Review #
- Test Run #

| Design Review # | Design Review Plan | Verification & Validation Plan | Test Run # | Test Issues | Test Issue Description |
|---|---|---|---|---|---|
| | | | | | |
| DR-2020.08.13-DR.001 | | | TR-2020.08.27-TR.001 | TR.001_2020.08.27_IS.001 | |
| DR-2020.08.13-DR.001 | | | TR-2020.08.27-TR.001 | | |

Launch Stage: Design Matrix
- Design Review #
- Test Run #

| Design Transfer | Design Transfer Confirmation | Design Change | Design Change Description |
|---|---|---|---|
| DT.001_2020.08.24_PRS.001 | | DC.001_2020.08.30_PRS.001 | |
| | | | |

***Design Matrix allows interaction between development teams while tracking QA, Test Issues, and Changes.***